

Unequal Job Opportunities and the Measurement of Welfare Inequality

Hisham Haydar

LISER and University of Luxembourg

France, EU-SILC priced through EUROMOD.

Seminar of 17 September. Welfare content is presented under ruling R6; the quantitative decomposition is work in progress.

Thank you. The question is how much of the inequality in money-metric well-being is attributable to unequal job opportunities rather than to differences in taste, and what happens to that attribution when a model leaves opportunities out. I will show the model, the estimates, the fit, the welfare definition and a descriptive welfare baseline. The quantitative decomposition is not ready and I will say exactly where it stands.

Question and contribution

Two people with the same tastes and the same wage need not face the same set of jobs.

- How much of measured welfare inequality is attributable to unequal **job access** and **earning opportunities**, rather than to **preferences**?
- What does a model that assumes a common choice set do to that attribution?

Both questions are about a *measure*, not about a policy counterfactual.

The two questions are the same two questions as in every version of this paper; nothing in the recent rulings touched them. The first is an attribution question about a money-metric inequality index. The second is a methodological question: a discrete-choice labour supply model that gives every household the same alternatives has to explain non-employment with a taste parameter, and I want to know whether that relabelling survives into the welfare accounting.

The contribution is a structural opportunity set plus a welfare measure defined on it.

- A random-utility, random-opportunity (RURO) model in which *which jobs a household can reach* is estimated, not assumed.
- The Haydar–Maniquet money-metric measure W^1 , applied on the estimated reachable set rather than on a common grid.
- A decomposition architecture that separates preferences from job access, earning opportunities and household endowments and needs.

Third item: the *architecture* is presented today. Its quantitative output is not.

The contribution has three parts and today only the first two carry numbers. The model estimates the opportunity distribution jointly with preferences. The welfare measure is taken from the companion theory paper with Francois Maniquet. The decomposition is the part under revision: the operators are settled, the estimand for the counterfactual attained bundle is not, and I will come back to that at the end.

Data and the structural RURO model

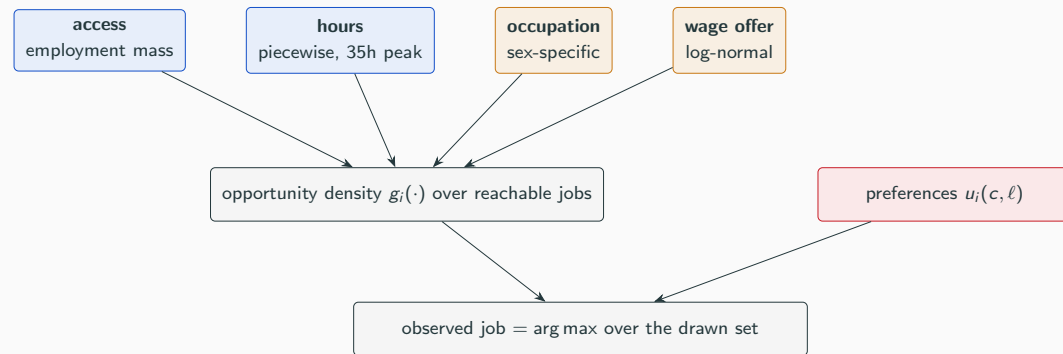
France, EU-SILC priced through EUROMOD: 1 540 single-adult and 2 223 couple households.

- Estimation frames: corrected criterion-A panels, S11 model of record.
- Each household: its observed job plus 100 drawn alternatives, every one priced through the tax–benefit system.
- Weights are the survey weights dwt throughout.

S11 is the single model of record on these slides. Earlier S8 and R240 frames and their estimates are not shown and are not comparable.

Two estimation samples, singles and couples, on the corrected criterion-A frames. Everything on the slides after this point is S11. I am deliberately not showing the older S8 or R240 numbers alongside, because the frames differ and a side-by-side would invite a comparison the samples do not support. Each household contributes its observed job and a hundred drawn alternatives, each of which is run through EUROMOD so the consumption argument is a real disposable income, not an approximation.

A job is a package; a household draws from its own distribution over packages.



Access and capability are not separately identified; the access factor is their product.

This is the architecture. A job is not an hours cell: it is hours, occupation and a wage, and whether the household can reach it at all. The opportunity density factorises into an access margin, a piecewise-constant hours factor with a peak at the statutory thirty-five hour week, sex-specific occupation availability, and an occupation-conditioned log-normal wage offer. Preferences enter separately. The observed job is the argmax over what was drawn. The identification caveat is on the slide and I keep it there all talk: the access factor mixes personal capability and market availability and the cross-section does not separate them.

Preferences and the opportunity density are estimated in one likelihood.

$$\Pr(j_i \mid x_i) \propto \exp\{u_i(c_{ij}, \ell_{ij}) + \log g_i(j) - \log q_i(j)\}$$

u preferences g estimated opportunity density q numerical proposal density, divided out

The proposal correction is what makes the drawn alternatives a simulator for the estimated set rather than the choice set itself.

One likelihood, two structural objects. The proposal density is a numerical device: the alternatives are drawn from it and it is divided back out, so it is not part of the economics. That separation matters later, because the welfare measure I am going to define must not depend on the proposal density, and I will show that it does not. Estimating preferences and opportunities jointly is what stops a taste parameter from quietly absorbing an availability parameter.

Positive estimates and fit

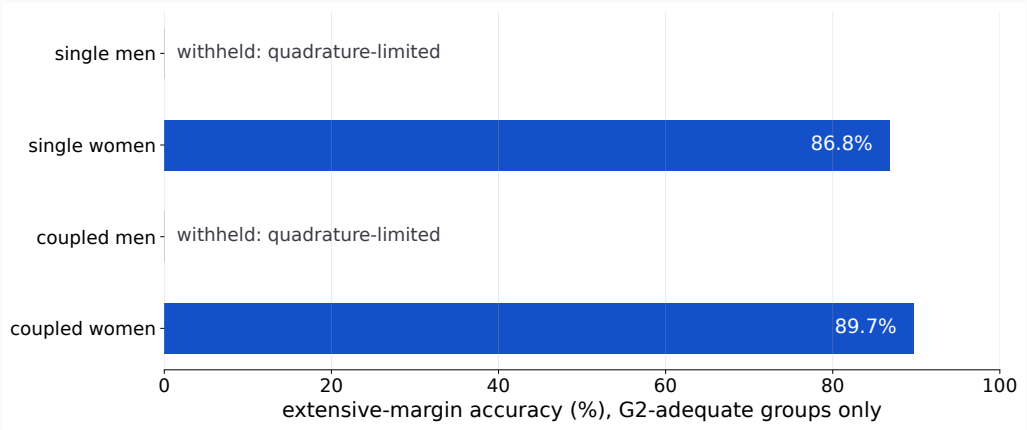
S11 model of record: 41 free parameters for singles, 47 for couples.

	singles	couples
free parameters	41	47
estimation households	1 540	2 223
objective at $\hat{\theta}$	6253.463	10283.034

Objectives reproduce exactly under the independent diagnostics evaluator, package commit 55bb0d0.

The headline fact about the estimates is that they are reproducible: an evaluator built separately for the diagnostics work recovers both stored objective values to the last digit shown. That is the precondition for everything on the next two slides. The parameter counts differ because the couples specification pins nothing at the bound while the singles specification pins eleven coefficients by construction.

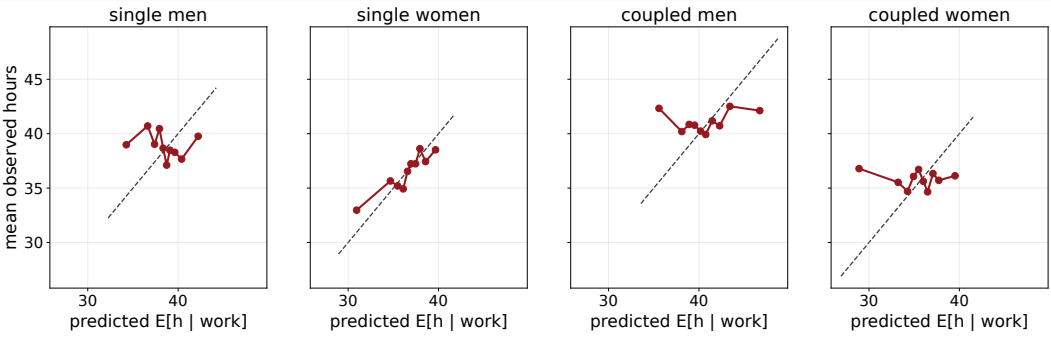
Extensive-margin accuracy is the one fit statistic that passes the quadrature-adequacy gate.



G2 gate: Monte Carlo s.e. $\leq 0.25 \times$ the sampling s.d. 2 of four groups pass; 2 are quadrature-limited and are withheld, not reported at a worse label.

I want to be explicit about what this slide does not say. The diagnostics run computes a long list of classification and calibration statistics. A pre-registered adequacy gate asks, for each one, whether the numerical integration error is small relative to the sampling variation the statistic would show anyway. Most of them fail that gate. I am showing the ones that pass and withholding the ones that do not, rather than showing them with a caveat attached, because a number that cannot be resolved by the quadrature is not a weaker finding, it is not a finding. Two of the four groups pass on extensive accuracy. The male groups do not, and their values are not on this slide for that reason.

Calibration conditioned on prediction: observed hours against predicted hours, by predicted decile.

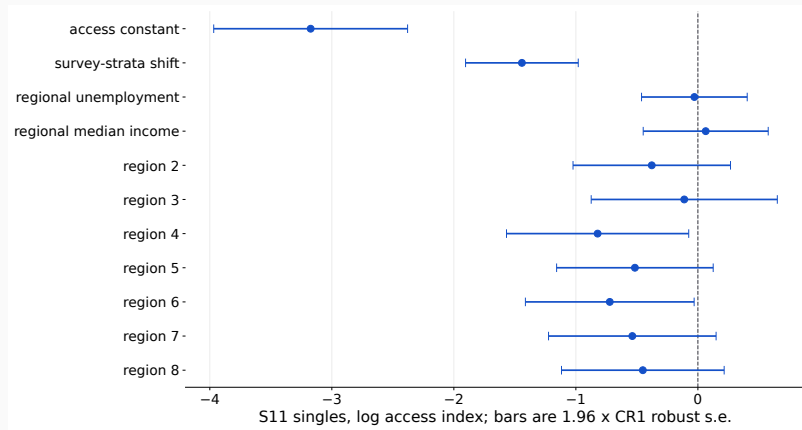


quadrature-limited; support audit pending

Households are sorted into deciles of their own predicted expected hours conditional on working, and each decile’s mean observed hours is plotted against its mean prediction. The dashed line is the forty-five degree line. Conditioning on the prediction rather than on the outcome is the direction that does not mechanically manufacture regression to the mean. The caption is the honest status: the statistics underlying this picture do not clear the quadrature gate, and the support audit that would settle whether the node set resolves the hours margin has not been done. So read the shape, not the gaps.

Evidence for heterogeneous opportunities

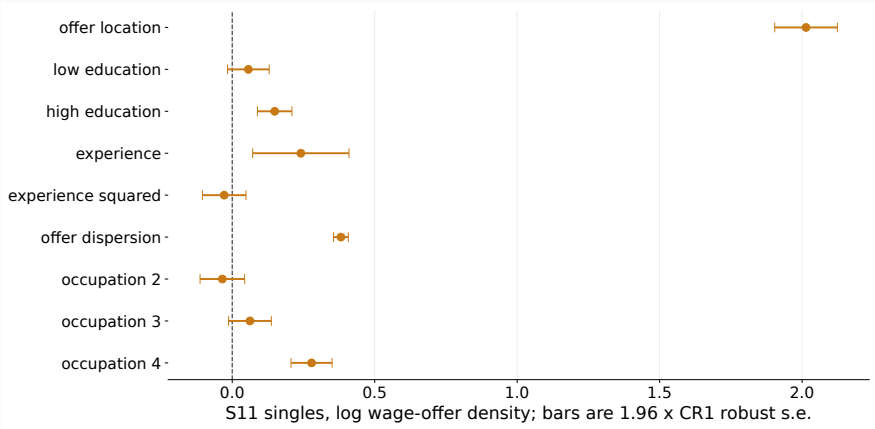
The access kernel is estimated, and it varies across households.



S11 singles; horizontal bars are $1.96 \times \text{CR1}$ cluster-robust standard errors. Structural and model-conditional; not causal.

This is the access block of the estimated opportunity density. The point is not any individual coefficient, it is that the block is estimated rather than imposed and that it moves: households in different regional access environments, and in different survey strata, face materially different employment masses at the same preferences and the same wage. This is the evidence that the opportunity side of the model is doing work. It is also the evidence the welfare measure later rests on, which is why I show it before the measure and not after.

So is the earning-opportunity kernel: an offer distribution, not a single wage.



S11 singles; occupation terms shift the offer location. Dispersion $\hat{\sigma} = 0.382$ ($z = 28.7$).

The wage side is an offer density with an estimated dispersion, not a point wage attached to each person. Education and experience shift the location, occupation shifts it again, and the residual dispersion is precisely estimated. Two households with identical observed wages can have different earning opportunities in this model, because what they face is a distribution. That is the second half of what I mean by unequal job opportunities, and together with the access block it is what the welfare measure has to be defined on.

The welfare measure

Haydar–Maniquet W^1 : the consumption at home that would leave the household exactly as well off.

$$W_i^1 = m_i(o; z_i) \quad \text{where} \quad u_i(m_i(o), o) = u_i(z_i)$$

z_i attained bundle o the non-employment state m_i the money metric

Mapping F is the primary empirical implementation (ruling R1). Mapping M is sensitivity only and requires an unidentified intensity parameter; it is not in the baseline.

Q&A note — ruling R1, JMP_W1_fork_ruling_v1.md Appendix A. If asked why this reference and not the market-job reference: the fork analysis compared Mapping F, the full feasible set including non-employment, against Mapping M, the market-job universe. R1 rules F primary for the current paper. M at its dense-law endpoint stops being a household-specific object and becomes a whole-market reference, and at finite intensity it needs a set-size parameter the data do not identify. That is recorded as a limitation and as future work, not as a competing headline. I am not claiming F is the right reference in the theory; I am claiming it is the one this empirical specification identifies.

Under the current specification W^1 coincides with the staying-home equivalent — as a fact about this domain.

$$W_{i,F}^1 = C_i^{\text{obs}} \exp \left[\frac{L_i(j_i^{\text{obs}}) - L_i(o)}{\beta_c} \right]$$

- Inputs: C_i^{obs} , $L_i(j_i^{\text{obs}})$, $L_i(o)$, β_c — nothing else.
- No opportunity density g , no proposal q , no shock ε , no intensity κ .

This is an **empirical-domain coincidence** under log consumption and universally available non-employment. It is **not** the theorem $W^1 = W^4$ in the Haydar–Maniquet theory.

Q&A note — ruling R1 and BASELINE-F-1 ruling section 2. The most likely question here is whether I have just proved that measure one and measure four are the same thing. I have not, and the ruling is explicit about it. The coincidence has two ingredients, both specific to what is estimated: consumption enters utility in logs, and non-employment is behaviourally available to every household. Change either and the two measures separate. If asked about the consumption argument: ruling E2 confirms that C^{obs} must be the exact object entering the estimated utility at the observed state, not a re-chosen normative income concept, and the measure-correspondence audit closed that provenance before the baseline was allowed to run.

**Equivalised W^1 -F distribution
(primary)**

Baseline W^1 -F, equivalised, single adults.

	C^{eq}	$W_F^{1\text{eq}}$
households	1 540	1 540
weighted mean	1 766	1 302
weighted median	1 588	1 165
weighted Gini	0.263292	0.249807

Units: EUR/month, household-equivalised (modified-OECD scale); survey weight dwt. A descriptive dispersion comparison, not a welfare-loss index. Equivalence scale: PROVISIONAL, pending economics review.

Source: docs/results/ JMP_BASELINE_F1_equivivalised_reporting_v1.md, commit 4c4e07e, over the verified construction (MNL commit 6048c9f, independently verified b5550af). This is a descriptive dispersion comparison (C^{eq} against $W_F^{1\text{eq}}$ within the singles sample), not a welfare-loss index.

Q&A note — BASELINE-F-1 ruling, E3-EQ, and ruling R6. Equivalised results are primary on this slide: the household-equivalised Gini of W_F^1 is below the household-equivalised Gini of observed consumption, for singles. I show that as a descriptive dispersion comparison only, not as a welfare-loss statement — the ratio W_F^1/C^{obs} reflects the systematic leisure term at the observed job relative to home leisure, not a monetary loss, and I am not constructing a loss index on this slide. The modified-OECD equivalence scale used here is flagged PROVISIONAL, pending an economics review that has not yet closed; I show the ratified units only, and I will say so if asked. The unequivalised construction this equivalises is unchanged and is in the backup appendix.

Baseline W^1 -F, equivalised, couples.

	C^{eq}	W_F^{1eq}
households	2 223	2 223
weighted mean	2 245	1 311
weighted median	2 063	1 237
weighted Gini	0.226805	0.197403

Units: EUR/month, household-equivalised (modified-OECD scale); survey weight dwt. A descriptive dispersion comparison, not a welfare-loss index. Equivalence scale: PROVISIONAL, pending economics review.

Reported separately from singles: no pooled figure, and no cross-sample level comparison is drawn on this slide or elsewhere in this deck. Source: same memo and commits as the singles equivalised slide.

Q&A note — BASELINE-F-1 ruling, E3-EQ, and ruling R6. Same descriptive comparison, on the couples sample. I am deliberately not placing this table next to the singles one: the two are separate estimation frames, the welfare unit differs, and R6 forbids presenting a singles/couples equivalised-level comparison as a finding. If asked how the couples numbers compare to singles, the honest answer is that this deck does not draw that comparison, and I would want to see it examined outside a seminar slide before treating it as informative. The scale-provisional caveat from the singles slide applies here too.

Decomposition architecture

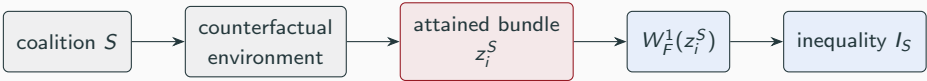
Four operators, one at a time: systematic utility heterogeneity, job access, earning opportunities, household endowments and needs.

operator	replaces, for every household
P	its systematic utility heterogeneity (tastes + reduced-form time constraints) by the common reference
A	its job access by the common reference access
B	its earning opportunities by the common reference offer distribution
D	its household endowments and needs by the common reference budget

Operators only. No share, no magnitude, no ordering is claimed on this slide or anywhere in this deck. Final decomposition semantics are work in progress.

Q&A note — ruling R2 and ruling R6. This is the architecture and deliberately nothing else. The earlier version of this talk put shares on this slide. Those shares were computed on an ex-ante inclusive-value functional, which the measure-correspondence audit established is a different object from Haydar–Maniquet measure one. R2 retires them from the baseline: they are preserved as historical research artifacts, they are not called measure one, and I am not presenting them as current results. If asked what the shares were, the answer is that I am not going to quote a number whose defining functional has been withdrawn. On P : I no longer call this operator a preference or taste parameter. Children enter the current structural specification through a reduced-form female time-constraint shifter. In a unitary labour-supply model this term may capture both tastes and childcare/home-production constraints; we therefore do not interpret it as pure preference heterogeneity. That is why P is labelled systematic utility heterogeneity, tastes plus reduced-form time constraints, rather than preferences alone, and why the final semantics of this decomposition are still work in progress.

The open problem: the baseline is evaluated at the observed bundle, but an operator moves it.



this step is the unresolved one

Two candidate estimands are under design (R5): a realised-bundle route with conditioned latent heterogeneity, and an ex-ante g -computation route. Neither is executed.

Q&A note — ruling R5. Here is the honest statement of where the paper is stuck. The welfare measure is evaluated at the bundle the household actually attained. Under a counterfactual that equalises access, the household would attain a different bundle, and the measure needs that bundle. Two routes are on the table. One conditions the behavioural latent state on the observed choice and carries it across counterfactual worlds, which raises the question of whether the max-stable construction supplies a legitimate cross-world coupling and whether it needs the unidentified intensity parameter. The other integrates the measure over the model-implied counterfactual choice distribution, which is cleaner but answers a different question: the inequality of expected welfare is not the expected inequality of welfare, and those two are not interchangeable. The design document has to pick one and defend it, and until it does there are no shares.

Work in progress

Where the quantitative welfare decomposition stands.

- **Settled.** The measure, its reference domain, the measure-correspondence audit, and the descriptive baseline just shown.
- **In design (R5).** The counterfactual attainment operator, and the choice between the two estimands.
- **Open.** The support audit behind the fit statistics, and coverage of the node set at the hours margin.
- **Withdrawn.** Every welfare share previously reported on this project's slides. They were computed on a different functional.

No P/A/B/D share is presented in this deck. R6 permits shares only once the revised F implementation has been executed and independently checked; it has not been.

Q&A note — ruling R6 and ruling R2. This slide is the replacement for what used to be the results section of this talk, and I would rather state it plainly than let it be discovered in questions. The welfare architecture is settled and the descriptive baseline is real and checked. The counterfactual machinery is in design. The old decomposition numbers are withdrawn, not revised: the object they were computed on is an ex-ante inclusive-value welfare functional, not the measure this paper now defines, so rescaling them would not repair them. What I am asking the audience for today is scrutiny of the measure and the opportunity kernel, which are the parts that are done.

What I would like from you today.

- Is Mapping F the reference you would want for this question?
- Which counterfactual attainment estimand would you defend?
- What would convince you the access kernel is identified, given that access and capability are not separated?

Thank you.

Q&A note — rulings R1, R5, and the identification caveat carried from the model slide. Three questions, in the order I most need answers. The reference domain choice is normative and I would like it challenged. The attainment estimand is the blocking design decision. And the access identification caveat is the one every discussant raises; I would rather hear what evidence would settle it than defend the current position.

households	1 540
weighted mean	1 435
weighted median	1 320
weighted Gini	0.238
observed workers	1 336
observed non-workers	204

Units: household EUR/month, unequivalised; survey weight dwt. Descriptive baseline; no counterfactual, no decomposition, no causal reading.

Secondary/verification slide: the equivalised singles slide (§6) is primary. Source: MNL/outputs/welfare/baseline_f1_v1/ baseline_f1_full_sample_report_v1.md, commit 6048c9f, independently verified b5550af.

This is the literal observed-bundle Mapping-F distribution before equivalisation, kept here so any question about the un-scaled level or about worker/non-worker counts has a one-slide answer. It is not the headline: the equivalised slide in §6 is. For non-workers the measure equals observed consumption exactly by construction; for workers it is strictly below it.

B2 — Baseline W^1 -F, unequivalised, couples (verification backup).

households	2 223
weighted mean	2 496
weighted median	2 356
weighted Gini	0.210
observed workers	2 173
observed non-workers	50

Units: household EUR/month, unequivalised; survey weight dwt. Descriptive baseline; no counterfactual, no decomposition, no causal reading.

Secondary/verification slide: the equivalised couples slide (§6) is primary. Reported separately from singles. No pooled figure: the two samples are separate estimation frames and the welfare unit differs.

The couples backup is on its own slide and there is no pooled row, because pooling would need an equivalence scale and the scale is a separate normative choice not yet ratified. These are unequivalised household totals: couples pool two earnings streams, and the non-employment count is small. The pricing-floor issue that affects the neither-work bundle for a subset of couples does not enter here, because this baseline is evaluated at the observed bundle and never prices the home state.

check	singles	couples
C1 non-workers, $\max W^1 - C^{\text{obs}} $	0	0
C2 workers, ratio outside $(0, 1)$	0	0
C3 dependency test $(g, q, \varepsilon, \kappa)$	PASS	PASS
C4 MEASURE-MAP-1R worked households	PASS	PASS
C5 indifference identity, $\max \Delta u $	1.8e-15	1.8e-15
C6 guards triggered	0	0

All six checks were pre-registered in the BASELINE-F-1 ruling before execution. C2 violations were to be explained, never clipped; none occurred.

The checks are the reason the baseline is presentable. C1 and C2 are the model-implied domain conditions. C3 is the dependency test: the baseline API accepts no opportunity density, no proposal, no shock and no intensity, and does not read them indirectly. C5 inverts the utility numerically and confirms the indifference identity to machine precision.

group	extensive accuracy	G2 label
coupled women	89.7%	ADEQUATE (ratio 0.17)
single women	86.8%	ADEQUATE (ratio 0.24)
coupled men	—	QUADRATURE-LIMITED, withheld
single men	—	QUADRATURE-LIMITED, withheld

Ratio is Monte Carlo s.e. over sampling s.d.; the gate is ≤ 0.25 . Node bootstrap: 200 replicates over 100 nodes. Simulated bands: 500 outcome vectors at fixed $\hat{\theta}$.

The gate was fixed before the statistics were computed. It asks whether numerical integration error is small enough relative to sampling variation for the statistic to be a statement about the model rather than about the quadrature. Withholding rather than downgrading is deliberate: a quadrature-limited number carries no information about fit.

ruling	what it settles here
R1	Mapping F primary; $W^1 = W^4$ is a domain coincidence
R2	W1-EA retired from the baseline; preserved, not repaired
R4	MEASURE-MAP-1R: the measure-correspondence gate
R5	counterfactual attainment: in design, not executed
R6	what this seminar may present
E1	BASELINE-F-1 authorised, executed, independently verified
E2	C^{obs} is the estimated utility's own argument
E3	C^{obs} benchmark: not authorised, not produced
E3-EQ	equivalised reporting (§6) is primary; scale PROVISIONAL

R1–R6: docs/normative/JMP_W1_fork_ruling_v1.md, Appendix A. E1–E3: docs/normative/JMP_W1_BASELINE_F1_authorization_and_Cobs_ruling_v1.md. E3-EQ: docs/results/JMP_BASELINE_F1_equivalised_reporting_v1.md, commit 4c4e07e.

This backup slide exists so that any question of the form “why is that not here” has a one-line answer with a document behind it.